

A SIMPLE EXAMPLE OF THE FAILURE OF THE NO-TRADE THEOREM DUE TO LACK OF COMMON KNOWLEDGE

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The purpose of this brief note is to provide a simple example of an environment in which the No-Trade theorem fails due to a lack of common knowledge. Specifically, you will see that there will be a state of the world in which it will be *mutually known* that trade is profitable. However, this fact will not be common knowledge and therefore agents will still wish to trade between themselves.[†]

1 Setup

Consider two agents. The first one has an indivisible good that she might be interested to sell to the second. This good can be sold tomorrow at an auction tomorrow. Let $(\Omega, 2^\Omega, \mu)$ be the initial finite probability space describing the uncertainty in this environment. Let T_1 , and T_2 be the type spaces and $\tau_1 : \Omega \rightarrow T_1$ and $\tau_2 : \Omega \rightarrow T_2$ be the private signaling functions of each of the agents. Let T denote the finite set of possible auction prices for the good. Let $P : \Omega \rightarrow T$ be the auction price.

We denote by Q_1 player 1's conditional expected price:

$$Q_1(\omega) = E[P | \Sigma_1](\omega),$$

Analogously,

$$Q_2(\omega) = E[P | \Sigma_2](\omega).$$

Theorem 1.1 (No-Trade Theorem, Milgrom and Stokey, 1982). *Let $(\Omega, 2^\Omega, \mu)$ be a probability space where μ is (1) a common prior. (2) If at some $\omega \in \Omega$ it is common knowledge that $Q_1 \leq q \leq Q_2$ for some q , then $Q_1(\omega) = q = Q_2(\omega)$.*

2 An example in which assumption (2) fails

Let $\Omega = \{\omega_1, \omega_2, \omega_3\}$, $T = \{L, H\}$, where H and L are two numbers such that $L < H$. Let $P : \Omega \rightarrow T$ be defined as $P(\omega_1) = P(\omega_3) = H$ and $P(\omega_2) = L$. Let $T_1 = \{a, b\}$ and $T_2 = \{a', b'\}$, and the signalling functions be defined as $\tau_1(\omega_1) = a$, $\tau_1(\omega_2) = \tau_1(\omega_3) = b$; $\tau_2(\omega_1) = \tau_2(\omega_2) = a'$, $\tau_2(\omega_3) = b'$. Let the common prior satisfy $\mu(\omega_1) = \mu_1$, $\mu(\omega_2) = \mu_2$, $\mu(\omega_3) = 1 - \mu_1 - \mu_2$, where the numbers μ_1 and μ_2 satisfy $0 < \mu_1 < 1$, $0 < \mu_2 < 1$, $0 < \mu_1 + \mu_2 < 1$, and $1 < 2\mu_1 + \mu_2$.

We have that Q_1 and Q_2 are given by

$$Q_1(\omega) = \begin{cases} H & \text{if } \omega = \omega_1 \\ \frac{(1-\mu_1)H - \mu_2(H-L)}{1-\mu_1} & \text{if } \omega \in \{\omega_2, \omega_3\} \end{cases} \quad Q_2(\omega) = \begin{cases} \frac{\mu_1 H + \mu_2 L}{\mu_1 + \mu_2} & \text{if } \omega \in \{\omega_1, \omega_2\} \\ H & \text{if } \omega = \omega_3 \end{cases}. \quad (1)$$

From the parametric assumptions above, it follows that $Q_1(\omega_1) > Q_2(\omega_2)$, $Q_1(\omega_2) < Q_2(\omega_2)$, and $Q_1(\omega_3) < Q_2(\omega_3)$. Thus, it follows that at state ω_1 , players 1 and 2 would not like to trade, whereas at states

[†]I would like to thank Abuzer Abuov, whose questions about the No-Trade theorem inspired me to construct the example in this note.

ω_2 and ω_3 they would like to trade. Formally, let $A \subset \Omega$ be the set of states at which *both* players would like to trade, i.e., $A = \{\omega \in \Omega \mid \exists q \text{ such that } Q_1(\omega) < q < Q_2(\omega)\}$. In light of the previous discussion, we have that $A = \{\omega_2, \omega_3\}$.

From the type spaces and the player's signalling functions we can compute the information partitions

$$\mathcal{P}_1 = \{\{\omega_1\}, \{\omega_2, \omega_3\}\}, \quad \mathcal{P}_2 = \{\{\omega_1, \omega_2\}, \{\omega_3\}\}.$$

Armed with the information partitions, we can easily analyze the knowledge the players have about this environment using the knowledge operator. Specifically, we are interested in what the players know about A and A^c , i.e., what they know about their mutual willingness or unwillingness to trade.

First note that $K_1 A = A$ and $K_2 A = \{\omega_3\}$. Thus, only at state ω_3 does player 1 know that player 2 is willing to trade and player 2 knows that player 1 is willing to trade. But what is the next level in the knowledge hierarchy? Note that $K_1 K_2 A = K_1 \{\omega_3\} = \emptyset$ and $K_2 K_1 A = K_2 A = \{\omega_3\}$. Thus even if the players *individually* know that the other is willing to trade at state ω_3 , Player 1 does not know that player 2 knows that player 1 is willing to trade at state ω_3 .

Exercise 1. What happens to knowledge about the event A^c ? What do you conclude.

Exercise 2. Construct an example in which both players know that the other knows that the other player wants to trade, but this fact is not known by at least one of the players in the next level of the knowledge hierarchy. Can you further generalize this example?

Exercise 3. Construct an example in which assumption (1) in the no-trade theorem fails, i.e., the player's don't have a common prior, but in which assumption (2) holds. Show that in your example the players would still be willing to trade.